

1

$$a) Y_t = 3 + Y_{t-1} + e_t - 0.75e_{t-1}$$

(1,0,0) = (1,0,0) $\rightarrow$ Autoregressive Moving average term

No differencing here, $d=0$

(Y_{t-1} term $\rightarrow p=1$)

($e_t - 0.75e_{t-1}$ term $\rightarrow q=1$)

$$\rightarrow (1, 0, 1)$$

As $d=0$, series is stationary

$$E(Y_{t+1}) = E(Y_t) = \mu$$

$$E(\Delta Y_t) = E(Y_t - Y_{t-1}) = 0$$

$$\text{Var}(Y_t - Y_{t-1}) = \text{Var}(3 + e_t - 0.75e_{t-1})$$

$$= \text{Var}(e_t - 0.75e_{t-1})$$

If endogenous constant variance of σ^2

$$\text{Var}(Y_t - Y_{t-1}) = 1^2 \text{Var}(e_t) + (-0.75)^2 \text{Var}(e_{t-1})$$

$$= 1 \cdot \sigma^2 + 0.5625 \sigma^2$$

$$= 1.5625 \sigma^2$$

noise is uncorrelated

$$b) Y_{t+1} = 10 + \underbrace{1.25Y_t + 0.25Y_{t-2}}_{AR} + \underbrace{e_t - 0.1e_{t-1}}_{MA}$$

here, $d=0$ (stationary)

$$p=2$$

$$q=10 - 10 + 1 + 1 + 8 = 10$$

$$(p, d, q) = (2, 0, 1)$$

$$E(\Delta Y_t) = E(Y_t - Y_{t-1}) \quad \text{similar to}$$

$$= 0 \quad \text{past @}$$

$$\text{Var}(\Delta Y_t) = \text{Var}(Y_t - Y_{t-1})$$

$$= \text{Var}(10 + 0.25 Y_{t-1} - 0.25 Y_{t-2} + e_t - 0.1 e_{t-1})$$

$$= \text{Var}(\underbrace{0.25 Y_{t-1} - 0.25 Y_{t-2}}_{\text{const}} + \underbrace{e_t - 0.1 e_{t-1}}_{\text{const}})$$

$$= \text{Var}(e_t + 0.1 e_{t-1}) \quad \text{stationary}$$

$$0 = (1 - 0.1) \sigma^2 = (0.9) \sigma^2$$

$$(1 - 0.25 - 0.1) \sigma^2 = (0.65) \sigma^2 = (0.65) \sigma^2$$

$$(1 - 0.25 - 0.1) \sigma^2 = \sigma^2 + (0.1)^2 \sigma^2$$

$$\sigma^2 = 0.1 \sigma^2$$

AR

$$Y_t = 10 + 0.25 Y_{t-1} - 0.25 Y_{t-2} + e_t - 0.1 e_{t-1}$$

$$Y_t = 10 + 0.25 Y_{t-1} - 0.25 Y_{t-2} + e_t - 0.1 e_{t-1}$$

MA

$p=3$ (three AR terms)

$d=0$ (no differencing)

$$q=2$$

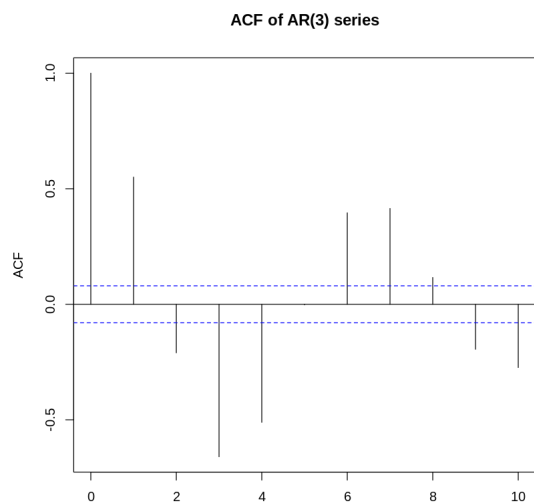
$$E(\Delta Y_t) = 0 \quad \text{(stationary)}$$

$$\begin{aligned}
 \text{Var}(\Delta Y_t) &= \text{Var}(Y_t - Y_{t-1}) \quad \nearrow \text{constants} \\
 &= \text{Var}(\underbrace{5 + Y_{t-1} - 1.7Y_{t-2} + 0.7Y_{t-3}}_{\text{const}} + e_t - 0.5e_{t-1} + 0.25e_{t-2}) \\
 &= \text{Var}(e_t - 0.5e_{t-1} + 0.25e_{t-2}) \\
 &= 1 \cdot \sigma^2 + (0.5)^2 \text{Var}(e_{t-1}) + (0.25)^2 \text{Var}(e_{t-2}) \\
 &= \sigma^2 + 0.25\sigma^2 + 0.0625\sigma^2 \\
 &= 1.3125\sigma^2
 \end{aligned}$$

```
#question 2
T <- 600
sigma <- sqrt(4)

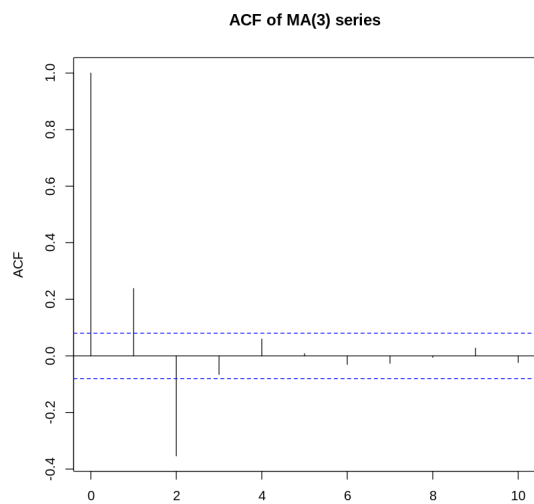
ar3 <- arima.sim(model = list(ar = c(0.8, -0.5, -0.2)), n = T, sd = sigma)

acf(ar3, lag.max = 10, main = "ACF of AR(3) series")
```



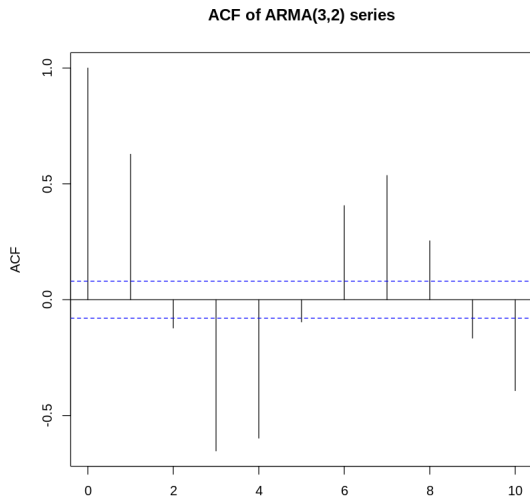
```
ma3 <- arima.sim(model = list(ma = c(0.8, -0.5, -0.2)), n = T, sd = sigma)

acf(ma3, lag.max = 10, main = "ACF of MA(3) series")
```



```
arma32 <- arima.sim(model = list(ar = c(0.8, -0.5, -0.2), ma = c(0.5, 0.3)), n = T, sd = sigma)

acf(arma32, lag.max = 10, main = "ACF of ARMA(3,2) series")
```



```
tacf_ar3 <- ARMAacf(ar = c(0.8, -0.5, -0.2), ma = numeric(0), lag.max = 10)
tacf_ar3
```



```
0:      1 1:      0.5555555555555556 2:      -0.1666666666666667 3:      -0.6111111111111111 4:      -0.5166666666666667 5:
```

for AR(3) model, the correlation is strong 0.555 and positive which is reflected in the sample acf plot, as the lag increase the correlation decreases to negative values and then it goes to positive again, alternating which is due to higher order of our AR model. sample acf is similar to our theoretical acf but varies slightly maybe due to noise.

```
tacf_ma3 <- ARMAacf(ar = numeric(0), ma = c(0.8, -0.5, -0.2), lag.max = 10)
tacf_ma3
```



for MA(3) model, we can observe the correlation is there initially but after 3 terms, it drops to zero in theoretical acf and our sample acf also has approximately zero correlation after three lag, the reason for slight variation could be due to noise.

```
tacf_arma32 <- ARMAacf(ar = c(0.8, -0.5, -0.2), ma = c(0.5, 0.3), lag.max = 10)
tacf_arma32
```



```
0:      1 1:      0.646942242355606 2:      -0.0583238958097395 3:      -0.570130237825594 4:      -0.556330690826727 5:
```

for ARMA(3,2) model, the theoretical acf values initially have high positive correlation but then go negative and then positive again they alternate, similar behaviour is showed by our sample acf plot, here also the difference at higher lags could be due to noise, especially if you see at lag 10

```
#question 3
install.packages("TSA")
library(TSA)
data(gold)
```



```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
also installing the dependencies 'xts', 'TTR', 'quadprog', 'zoo', 'quantmod', 'leaps', 'locfit', 'tseries'
```

```
Attaching package: 'TSA'
```

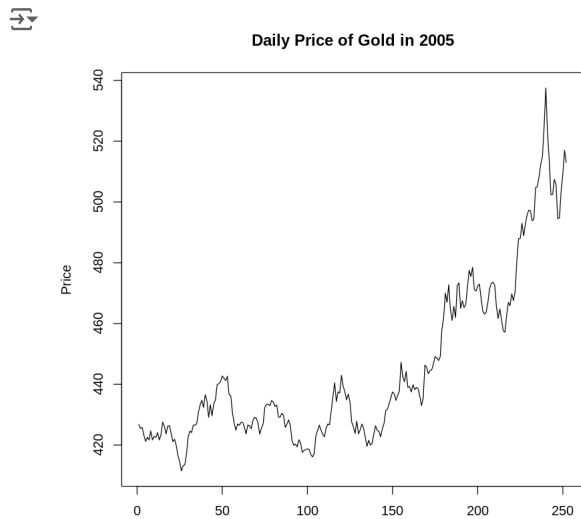
```
The following objects are masked from 'package:stats':
```

```
acf, arima
```

```
The following object is masked from 'package:utils':
```

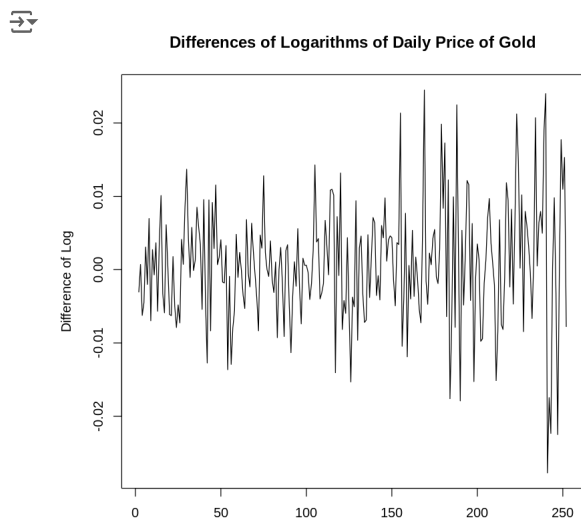
```
tar
```

```
plot(gold, type = "l", main = "Daily Price of Gold in 2005", ylab = "Price", xlab = "Day")
```



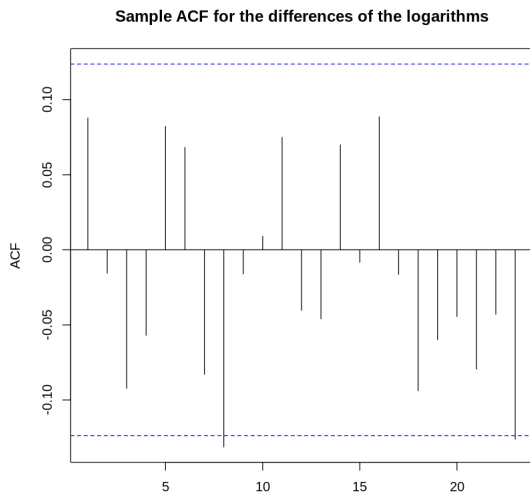
it shows overall upward trend, near 150 the price raises steadily, initially it fluctuates a little but towards the end the price is significantly higher than the start.

```
log_diff_gold <- diff(log(gold))  
plot(log_diff_gold, type = "l", main = "Differences of Logarithms of Daily Price of Gold", ylab = "Difference of Log", xlab =
```



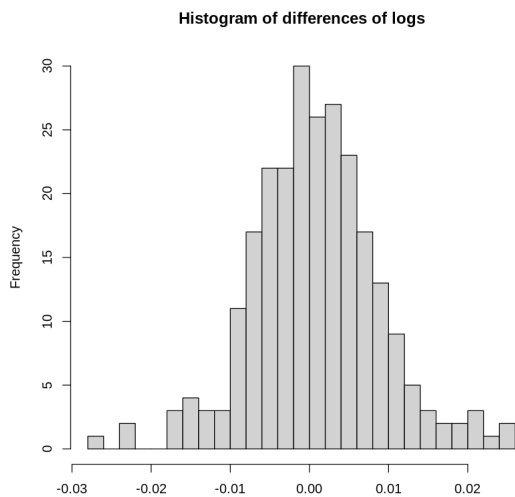
this is clearly a time series, initially they are in smaler range towards the end the difference became much high in magnitude which indicates the significant spike in price to gold towards the end.

```
acf(log_diff_gold, main = "Sample ACF for the differences of the logarithms")
```



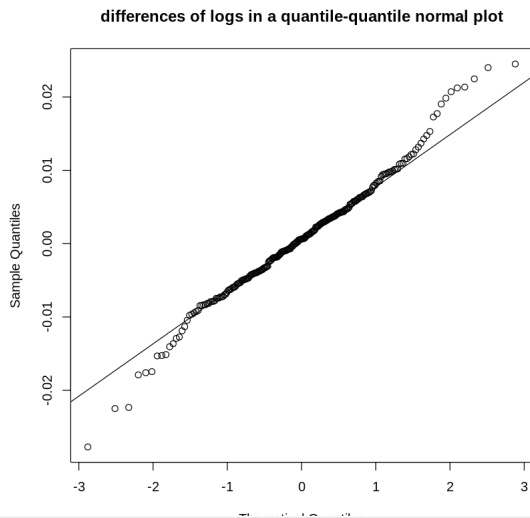
here, the correlations donot seem to follow any order, clearly saying the differenced series is uncorrelated, even though there are some that are outside confidence interval, it's very minimal and could be due to noise, so we can conclude that differenced series is stationary and this is random walk model

```
hist(log_diff_gold, breaks = 20, main = "Histogram of differences of logs", xlab = "Log Difference")
```



clearly we see a peak around zero log difference, and it's shape is approximately symmetric around zero and similar to normal distribution, and this is symmetry shows that there is no trend, and this is consistent with our conclusion from part c

```
qqnorm(log_diff_gold, main = "differences of logs in a quantile-quantile normal plot")
qqline(log_diff_gold)
```



most points are on the line and only few deviate at both ends, data seems to be centered around zero, so this suggests that the difference in logs are normally distributed.

```
#question 4
install.packages("lubridate")
library(lubridate)

data(winnebago)
time <- seq(from = as.Date("01/11/1966", format="%d/%m/%Y"),
            to = as.Date("01/02/1972", format="%d/%m/%Y"), by = "month")
quart <- quarters(time)
month.vals <- as.factor(month(time))
```



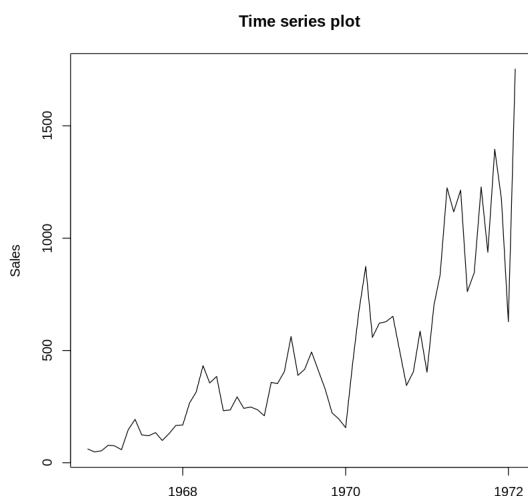
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
plot(time, winnebago, type = "l", main = "Time series plot ", ylab = "Sales", xlab = "Time")
```



the sales clearly increase over years, showing an upward trend. there is a sharp increase towards the end around year 1972, there are fluctuations some small some large, the big rise and falls can be seen at some years.

```
time_index <- 1:length(winnebago)
fit_linear <- lm(winnebago ~ time_index)
```

```
summary(fit_linear)
```

```
std_residuals <- rstandard(fit_linear)
plot(time, std_residuals, type = "l", main = "Standardized Residuals", ylab = "Standardized Residuals", xlab = "Time")
abline(h = 0, col = "red")
```

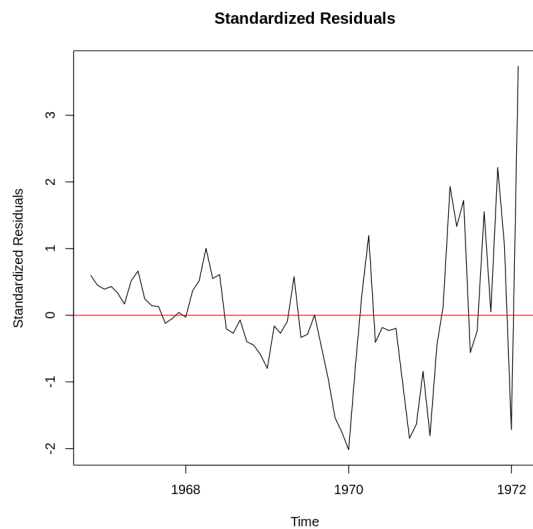


```
Call:
lm(formula = winnebago ~ time_index)

Residuals:
    Min       1Q   Median       3Q      Max
-419.58  -93.13  -12.78   94.96  759.21

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -76.832     53.053  -1.448   0.153
time_index    16.728      1.419   11.788 <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

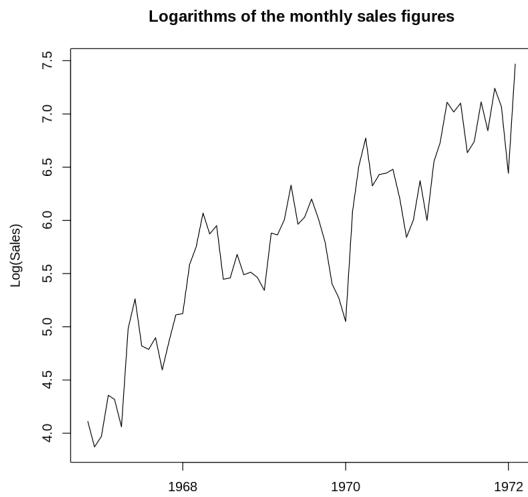
Residual standard error: 209.7 on 62 degrees of freedom
Multiple R-squared:  0.6915,    Adjusted R-squared:  0.6865
F-statistic: 138.9 on 1 and 62 DF,  p-value: < 2.2e-16
```



time index slope coeff is 16.728 showing sales increase by 16.728 per month, the p value is small meaning increase is significant, R squared is 0.6915 meaning 69.15% variability in sales can be explained by our linear trend. RSS is 209.7 suggest variability not captured by the trend, f-statistic and p value suggest model is good.

the standard residual values fluctuate around zero, the range of this fluctuations increase over time, some are fluctuating more indicating outliers this is due to that spike in our time series around 1972.

```
log_winnebago <- log(winnebago)
plot(time, log_winnebago, type = "l", main = "Logarithms of the monthly sales figures", ylab = "Log(Sales)", xlab = "Time")
```



it shows an upward trend, there are some fluctuations and they look similar could be due to periodic nature of the sales (like naturally sales peak around december and go down in summer something like that), and we see a big straight jump around 1970 and around 1972 meaning growth in sales

```
fit_log_linear <- lm(log_winnebago ~ time_index)
```

```
summary(fit_log_linear)
```

```
std_residuals_log <- rstandard(fit_log_linear)
```

```
plot(time, std_residuals_log, type = "l", main = "Standardized Residuals of logged data", ylab = "Standardized Residuals", xlab = "Time", abline(h = 0, col = "red"))
```



Call:

```
lm(formula = log_winnebago ~ time_index)
```

Residuals:

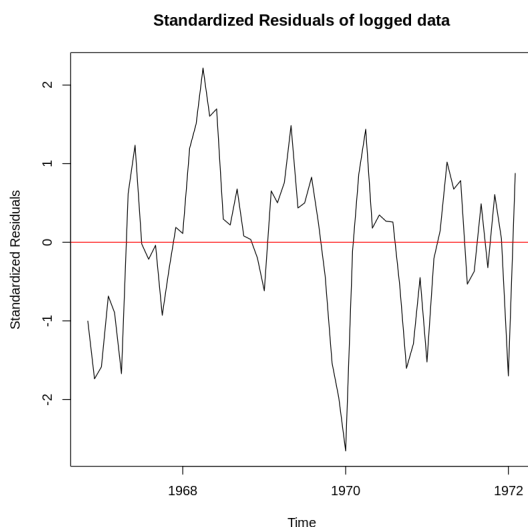
Min	1Q	Median	3Q	Max
-1.03669	-0.20823	0.04995	0.25662	0.86223

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.451608	0.099644	44.67	<2e-16 ***
time_index	0.041922	0.002665	15.73	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3939 on 62 degrees of freedom
Multiple R-squared: 0.7996, Adjusted R-squared: 0.7964
F-statistic: 247.4 on 1 and 62 DF, p-value: < 2.2e-16



time index slope coeff is 0.041 showing sales increase by $e^{(0.041)}$ per month, the p value is small meaning values of intercept and time index slope is significant, R squared is 0.7996 about 79.9% variance is logged data is explained. RSS is 0.3939, f-statistic and p value suggest model is good.

the standard residual values fluctuate around zero, the range of this fluctuations doesn't really have a trend, some are fluctuating more indicating outliers this is due to that spike in our time series around 1972.

```
fit_seasonal <- lm(log_winnebago ~ time_index + month.vals)
summary(fit_seasonal)

std_residuals_seasonal <- rstandard(fit_seasonal)

plot(time, std_residuals_seasonal, type = "l",
     main = "Standardized Residuals from Seasonal Model",
     ylab = "Standardized Residuals", xlab = "Time")
abline(h = 0, col = "red")
```

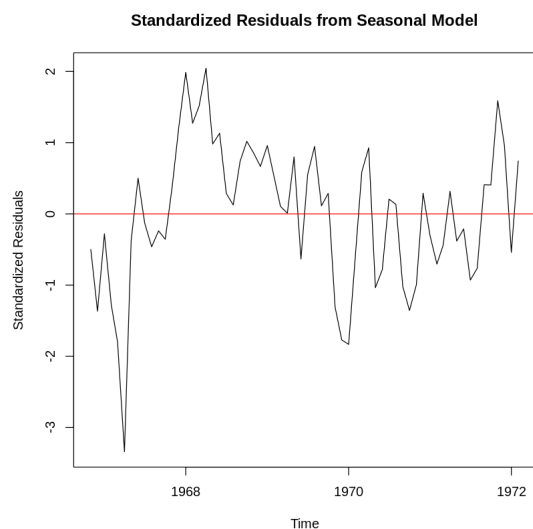


```
Call:
lm(formula = log_winnebago ~ time_index + month.vals)

Residuals:
    Min       1Q   Median       3Q      Max
-0.92501 -0.16328  0.03344  0.20757  0.57388

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  3.921324   0.146722  26.726 < 2e-16 ***
time_index    0.042424   0.002143  19.800 < 2e-16 ***
month.vals2    0.624448   0.181824   3.434 0.001188 **
month.vals3    0.682197   0.190878   3.574 0.000779 ***
month.vals4    0.809588   0.190794   4.243 9.30e-05 ***
month.vals5    0.869525   0.190733   4.559 3.25e-05 ***
month.vals6    0.863087   0.190697   4.526 3.63e-05 ***
month.vals7    0.553918   0.190685   2.905 0.005420 **
month.vals8    0.569885   0.190697   2.988 0.004305 **
month.vals9    0.575717   0.190733   3.018 0.003960 **
month.vals10   0.263486   0.190794   1.381 0.173300
month.vals11   0.286822   0.181862   1.577 0.120946
month.vals12   0.248022   0.181824   1.364 0.178532
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.3149 on 51 degrees of freedom
Multiple R-squared:  0.8946,    Adjusted R-squared:  0.8699
F-statistic: 36.09 on 12 and 51 DF,  p-value: < 2.2e-16
```



time index slope coeff is 0.0424 showing sales increase by $e^{(0.0424)}$ per month, the p value is small meaning increase is significant, R squared is 0.8969 meaning 89.69% variability in logged sales can be explained by our linear trend. RSS is 0.3149 suggest better fit, f-statistic and p value suggest model is good.

```
#question5
install.packages("reshape2")
library(reshape2)
```

```

Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)

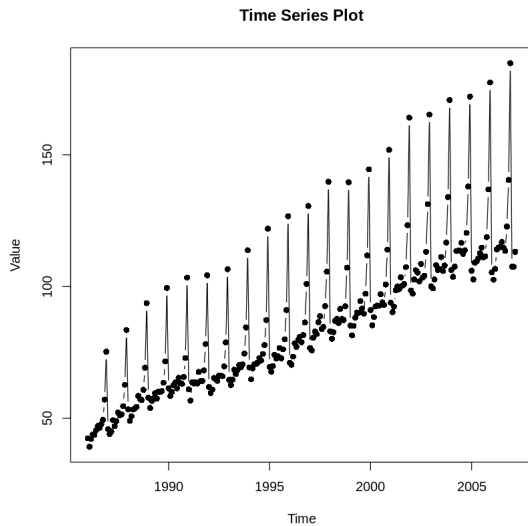
```

```

data(retail)
retail.long <- melt(retail)
time <- seq(from=as.Date("01/01/1986", format="%d/%m/%Y"),
            to=as.Date("01/03/2007", format="%d/%m/%Y"), by="month")
quart <- quarters(time)
month.vals <- as.factor(month(time))

plot(time, retail.long$value, type="b", pch=16, main="Time Series Plot",
     xlab="Time", ylab="Value")

```



It clearly has an upward trend, retail sales are going up with time. It shows repeated patterns every year showing seasonality, the fluctuations appear to increase with time, meaning the seasonal variation is also growing along with the retail sales.

```

retail.long$month <- as.factor(month(time))

model <- lm(value ~ time + month, data=retail.long)

summary(model)

std_residuals <- residuals(model) / sd(residuals(model))

```

Call:
lm(formula = value ~ time + month, data = retail.long)

Residuals:

	Min	1Q	Median	3Q	Max
	-19.8940	-2.4441	-0.3532	2.1969	16.2059

Coefficients:

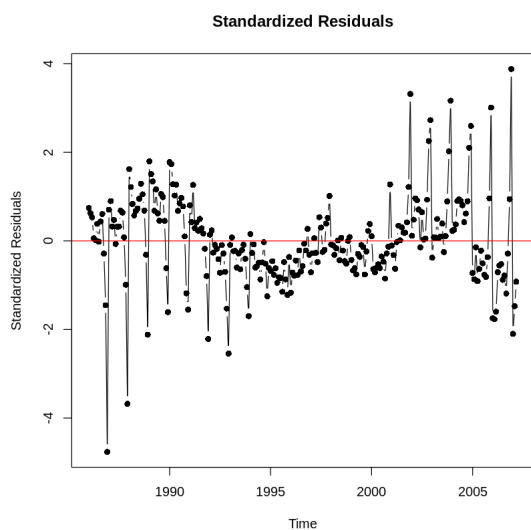
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-1.943e+01	1.474e+00	-13.184	< 2e-16 ***
time	1.005e-02	1.196e-04	83.998	< 2e-16 ***
month2	-3.021e+00	1.290e+00	-2.342	0.02000 *
month3	9.125e-02	1.290e+00	0.071	0.94366
month4	3.458e+00	1.305e+00	2.649	0.00859 **
month5	3.123e+00	1.305e+00	2.393	0.01748 *
month6	3.083e+00	1.305e+00	2.362	0.01895 *
month7	6.067e+00	1.305e+00	4.649	5.49e-06 ***
month8	3.146e+00	1.305e+00	2.411	0.01666 *
month9	3.430e+00	1.305e+00	2.628	0.00913 **
month10	8.562e+00	1.305e+00	6.560	3.23e-10 ***
month11	2.082e+01	1.305e+00	15.950	< 2e-16 ***
month12	5.255e+01	1.305e+00	40.261	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.278 on 242 degrees of freedom
Multiple R-squared: 0.9767, Adjusted R-squared: 0.9755
F-statistic: 845 on 12 and 242 DF, p-value: < 2.2e-16

time index slope coeff is $1.005e-02$ showing sales increase by $e^{(1.005e-02)}$ per month, the p value is small meaning values of intercept and time index slope is significant. R squared is 0.9767 about 97.67% variance is data is explained. RSS is 4.278. f-statistic and p value suggest

```
plot(time, std_residuals, type="b", pch=16, main="Standardized Residuals",
      xlab="Time", ylab="Standardized Residuals")
abline(h=0, col="red")
```



the standard residual values fluctuate around zero, the range of this fluctuations is similar over time, some are fluctuating more indicating outliers, this reflect time series plot. this means that model fits most data but there is room for making it better.